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Inverse mass cascade in dark matter flow and effects on halo deformation, energy, size, and density profiles

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ABSTRACT

Dark matter haloes grow by accretion, and the inverse mass cascade sets how fast. We ask what continuous growth implies for the internal motion, energy and density profile of a halo. If a halo grows self-similarly, with mass and radius both proportional to time at fixed concentration c, mass conservation fixes the mean radial velocity inside it for any density profile: outward in the core and inward beyond about twice the scale radius. For an NFW halo the net radial momentum vanishes at $c_0 = 3.48$. Growth adds surface and kinetic terms to the virial balance; at c_0 they give an effective virial exponent $n_e = -1.42$ instead of -1. A halo radius that grows by random accretion events follows geometric Brownian motion, and particles in such a halo obey a Fokker–Planck equation whose solution, for a simple osmotic flow, is a generalized gamma profile that contains the Einasto profile as a special case. We test these results on composite haloes (all haloes with the same particle number, 4–6326 particles) of a 256^3 SCDM simulation. The generalized gamma profile fits their cumulative mass profiles with a median rms residual of 0.006 dex and $a_r/b_r = 1.41 \pm 0.05$; the flat inner part lies within about one softening length. The measured mean radial velocity, however, is inward at all resolved radii, between $-0.39\,Hr$ and $-1.24\,Hr$, where self-similar growth predicts about $+0.53\,Hr$ at the scale radius, and the radial velocity dispersion exceeds the isotropic Jeans value by 1.11–1.90 times. Self-similar growth describes the profile shapes, but not the instantaneous kinematics, of these haloes.

Key words: dark matter – methods: numerical – galaxies: haloes

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1 INTRODUCTION

Dark matter haloes are the gravitationally bound structures in which galaxies and galaxy clusters form, and their density profiles connect the dark matter model to almost every observation of it. Cosmological N-body simulations find nearly universal profiles, shallower than an isothermal sphere at small radius and steeper at large radius [1, 2]; the Einasto profile, whose logarithmic slope varies as a power of radius, describes them better than a fixed inner cusp [3, 4, 5, 6], and the same form holds over twenty orders of magnitude in halo mass [7]. The profile and its concentration set the rotation curves of galaxies [8, 9], the lensing masses of galaxy clusters [10] and the expected signals of dark matter annihilation [11], and the inner slope is at the centre of the cusp–core problem of dwarf and low-surface-brightness galaxies [12, 13]. The concentration–mass relation is therefore a standard ingredient of halo models and of lensing and cluster analyses [14, 15, 16, 17].

Haloes are never finished: they keep merging and accreting, and their structure records how they grew. The concentration of a halo is set by its mass accretion history [18, 19, 20, 21, 22], and a fast early phase of growth followed by a slow phase has been proposed as the origin of the profile shape [23]. Self-similar infall models link growth and structure analytically [24, 25, 26]. The outer profile depends on the accretion rate [27], and the splashback radius that marks the edge of the orbiting matter [28] has been detected in the galaxy distribution around clusters [29]. Haloes are thus not static: matter still falls in around and inside the region usually called the virial radius [30, 31], and dynamical mass estimates must allow for the surface pressure at the halo edge [32]. Yet the internal mean velocity of a growing halo, and its effect on the dispersion, the virial balance and the profile, is usually left out of equilibrium models of haloes.

A statistical description of halo growth comes from an analogy with turbulence. In three-dimensional turbulence, energy is passed from large eddies to small ones [33], and in two-dimensional turbulence it is passed the other way, from small to large scales [34]. In collisionless dark matter flow, whose elementary structures are haloes [35, 36], mass is passed from small haloes to large ones. This inverse mass cascade has a scale-independent mass flux in a propagation range of halo masses and sets the halo mass function [37, 38];

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Table 1. Parameters of the simulation.

Run	Ω_0	Λ	h	Γ	σ_8	L (Mpc/h)	N	m_p ($M_\odot/h$)	$l_{\rm soft}$ (kpc/h)
SCDM1	1.0	0.0	0.5	0.5	0.51	239.5	256^3	2.27×10^{11}	36

the energy carried along with the mass forms an inverse energy cascade [39], and random walks of halo mass and size lead to the halo mass functions and density profiles [40, 41]. The cascade gives a growth law for haloes. The question of this paper is what that continuous growth does to the inside of a halo.

We follow a single kinematic idea: if a halo keeps its shape while its mass and size grow, mass conservation fixes how the matter inside it must move. The answer is a mean radial flow, and from it follow a scale radius, a preferred concentration, extra surface and kinetic terms in the virial balance and, when growth is random, a stochastic model of the density profile. For an NFW halo the net radial momentum of the flow vanishes at $c_0 = 3.48$, and the effective virial exponent there is $n_e = -1.42$ instead of -1; the stochastic model gives a generalized gamma profile that contains the Einasto profile. Because the idea is simple, it can be tested. We measure the cumulative mass profiles, the mean radial velocity and the velocity dispersion of composite haloes of 4–6326 particles in a cosmological simulation. The profile model fits them with a median rms residual of 0.006 dex and $a_r/b_r = 1.41\pm0.05$. The radial flow, however, does not have the predicted sign: it is inward at all resolved radii, between $-0.39\,Hr$ and $-1.24\,Hr$, and the radial dispersion exceeds the isotropic Jeans value by 1.11–1.90 times. Both results are reported.

Section 2 describes the simulation and the composite haloes. Section 3 derives the radial flow of a growing halo, Section 4 its momentum, dispersion and energy, and Section 5 the stochastic models of halo size and particle distribution. Section 6 compares the predictions with the simulation, and Sections 7 and 8 discuss and summarize the results.

2 SIMULATION AND COMPOSITE HALOES

We use the SCDM run of the Virgo consortium [42, 43], which has been used for clustering statistics [43], cluster formation [44] and mass functions [45]. Its parameters are listed in Table 1. The universe is matter dominated, so $H = 2/(3t)$ and $a \propto t^{2/3}$ throughout. Haloes were identified by friends-of-friends with a linking length of 0.2 Mpc/h (comoving). Halo-based statistics of this run are public [46, 47, 48].

Composite haloes. Individual haloes with tens of particles are too noisy for a profile. We therefore stack all haloes with the same number of particles n_p about their own centres of mass and call the stack a composite halo. At $z = 0$ the 1,674,583 haloes with $n_p \geqslant 2$ contain 10,290,483 particles in 1092 groups, from $n_p = 2$ (about 956,000 pairs) to $n_p = 6326$ (one halo). For every particle we use its distance r to the centre of mass of its own halo, its peculiar radial velocity u_r and its two tangential velocity components relative to the halo. The cumulative mass fraction of a composite halo is

$$F_r(r) = \frac{N(<r)}{n_p\, n_h}, \tag{1}$$

where $N(<r)$ is the number of particles of the group closer than r to the centre of their halo and n_h is the number of haloes in the group: $F_r(r)$ is the fraction of the mass of an average halo inside radius r. Stacking reduces the noise because a group holds many more particles than any one of its haloes. At $z = 0$ comoving and physical distances are equal; the Hubble flow is Hr with $H_0 = 100\,h\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Two scales limit what the data can show. The softening length, $l_{\rm soft} = 0.036$ Mpc/h, sets the smallest resolved radius; we treat $r < 2\,l_{\rm soft}$ as unresolved [49]. A composite halo also mixes haloes of different shapes and histories, so its profile describes the ensemble, not a single object.

3 KINEMATICS OF A GROWING HALO

3.1 Growth from the mass cascade

Take a halo of mass m_h and virial radius r_h whose mean density is $\Delta_c = 18\pi^2$ times the background density $\bar\rho = \bar\rho_0 a^{-3}$:

$$m_h = \frac{4}{3}\pi r_h^3 \Delta_c \bar\rho_0 a^{-3}. \tag{2}$$

With $H^2a^3 = 8\pi G\bar\rho_0/3$ the circular velocity at the halo surface is

$$v_{\rm cir}^2 = \frac{Gm_h}{r_h} = (3\pi H r_h)^2 = \frac{4\pi^2 r_h^2}{t^2}. \tag{3}$$

A halo is therefore a structure whose edge rotates once per Hubble time, $2\pi r_h/v_{\rm cir} = t$.

In the inverse mass cascade a halo of $n_p = m_h/m_p$ particles waits a time τ_g before it gains (or passes on) one particle mass m_p, so $\mathrm{d}m_h/\mathrm{d}t = m_p/\tau_g$ [37]. The time needed to assemble the whole halo, $\tau_f = \tau_g n_p$, is proportional to t, with $\tau_f H = 2(1-\lambda)/(3-2\tau_0)$, where λ and τ_0 are the two exponents of the cascade (the waiting time scales as $\tau_g \propto m_h^{-\lambda} a^{\tau_0}$). With equation (2) this gives the growth rates of mass and size,

$$s_m \equiv \frac{\partial \ln m_h}{\partial \ln a} = \frac{3-2\tau_0}{2(1-\lambda)}, \qquad s_r \equiv \frac{\partial \ln r_h}{\partial \ln a} = \frac{9-2\tau_0-6\lambda}{6(1-\lambda)}, \tag{4}$$

so that $s_r = s_m/3 + 1$. For $\lambda = 2/3$ and $\tau_0 = 1$ both are $3/2$: mass and radius grow in proportion to t, and $\partial \ln m_h/\partial \ln r_h = 1$. We call this *self-similar growth* and use it in the rest of the paper unless stated otherwise.

3.2 Surface density and the deformation parameter

During a short time $\mathrm{d}t$ the halo gains a shell of N_s particles of thickness r_p at its surface, where the density is $\rho_h(r_h) = N_s m_p/(4\pi r_h^2 r_p)$ (Fig. 1). The new mass also pulls on the old halo, whose surface moves inward by r_p'. The radius therefore grows by $\mathrm{d}r_h = r_p - r_p' = \alpha_h r_p$, which defines the deformation parameter

$$\alpha_h = 1 - \frac{r_p'}{r_p}. \tag{5}$$

An undeformed halo has $\alpha_h = 1$. Equating the growth of r_h from the shell with the growth from equation (4) gives the surface density

$$\rho_h(r_h) = \frac{m_h}{4\pi r_h^3}\,\frac{3\alpha_h}{1+3\tau_g H n_p} = \frac{\alpha_h m_h}{4\pi r_h^3}, \tag{6}$$

where the second form uses $\tau_g H n_p = 2/3$ ($\lambda = 2/3$, $\tau_0 = 1$). The surface density is $\alpha_h/3$ of the mean density $\bar\rho_h = 3m_h/(4\pi r_h^3)$. For a power-law profile $\rho_h \propto r^{-m}$ the surface density is $(3-m)m_h/(4\pi r_h^3)$ (Appendix A), so $m = 3-\alpha_h$: an undeformed halo is isothermal ($m = 2$), and the steepest possible power law, $m = 3$, corresponds to $\alpha_h = 0$.

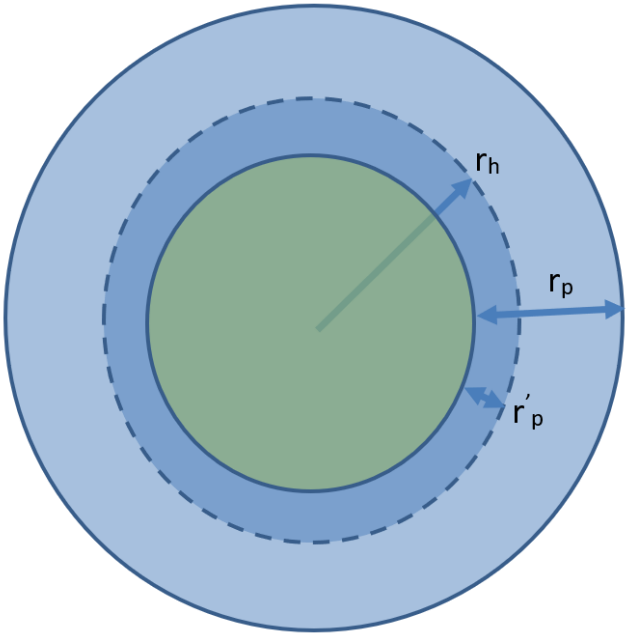


Figure 1. Growth of a halo during a time dt. The halo of radius r_h (dashed) gains a shell of mass of thickness r_p, and the old halo is compressed by r'_p (green). The radius grows by $r_p - r'_p = \alpha_h r_p$. With $r'_p = 0$ the halo is not deformed and its profile is isothermal.

For a general profile write the enclosed mass as

$$m_r(r,t) = m_h \frac{F(x)}{F(c)}, \qquad x = \frac{r}{r_s} = \frac{cr}{r_h}, \tag{7}$$

where r_s is the scale radius, $c = r_h/r_s$ the concentration and F a cumulative shape function with $F(0) = 0$: $F = x$ for the isothermal sphere, $F = \ln(1+x) - x/(1+x)$ for NFW and an incomplete gamma function for Einasto (Appendix A). The density is

$$\rho_h = \frac{1}{4\pi r^2}\frac{\partial m_r}{\partial r} = \frac{m_h}{4\pi r_h^3}\frac{c^3 F'(x)}{x^2 F(c)}, \qquad \frac{\partial \ln \rho_h}{\partial \ln x} = \frac{xF''}{F'} - 2. \tag{8}$$

Comparing with equation (6) at $x = c$ identifies the deformation parameter with the logarithmic slope of the enclosed mass at the edge,

$$\alpha_h = \frac{cF'(c)}{F(c)}. \tag{9}$$

For NFW, $\alpha_h = 0.79$ at $c = 4$ and 0.56 at $c = 10$; $\alpha_h = 1$ at $c = 2.16$. Less concentrated (more massive) haloes are less deformed. Equivalently, $\bar\rho_h/\rho_h(r_h) = 3F(c)(1+1/c)^2 = 3/\alpha_h$ for NFW, a relation between concentration and deformation.

3.3 Mean radial flow

A halo that keeps the form (7) while m_h, r_s and c change cannot be static inside. The continuity equation, $\partial_t \rho_h + r^{-2}\partial_r(r^2\rho_h u_r) = 0$, integrates to $\partial_t m_r = -4\pi r^2 \rho_h u_r$. Inserting equation (7) gives the mean radial velocity

$$u_r = \frac{r_s}{t}\left[x\frac{\partial \ln r_s}{\partial \ln t} + \left(\frac{\partial \ln F(c)}{\partial \ln t} - \frac{\partial \ln m_h}{\partial \ln t}\right)\frac{F(x)}{F'(x)}\right]. \tag{10}$$

The first term stretches the profile with the scale radius; the second removes the mass that the growth of m_h adds. At constant c, and in units of the core expansion speed r_s/t,

$$u_h(x) \equiv \frac{u_r}{r_s/t} = \left[x - \frac{\partial \ln m_h}{\partial \ln r_h}\frac{F(x)}{F'(x)}\right]\frac{\partial \ln r_h}{\partial \ln t} = x - \frac{F(x)}{F'(x)}, \tag{11}$$

where the last form holds for self-similar growth. Equation (11) is the central kinematic result: the shape of the profile alone fixes the mean flow. Its properties follow from $\mathrm{d}(F/F')/\mathrm{d}x = 1 - FF''/F'^2$:

- $u_h(0) = 0$, and near the centre $u_h \simeq \gamma_h x$ with the deformation rate $\gamma_h = u'_h(0)$; $\gamma_h = 1/2$ for NFW and $2/3$ for Einasto.
- $u'_h = FF''/F'^2$ vanishes where $F'' = 0$, i.e. where $\partial \ln \rho_h/\partial \ln x = -2$. For NFW and Einasto this is $x = 1$: the flow is fastest at the scale radius.
- $u_h = 0$ where $F/F' = x$, which is also where the circular velocity $v_c^2 \propto F(x)/x$ peaks ($x_0 \approx 2.16$ for NFW).
- At the edge, $u_h(c) = c(1 - 1/\alpha_h)$: inflow when $\alpha_h < 1$.
- The isothermal sphere ($F = x$) has $u_h = 0$: no deformation, no flow.

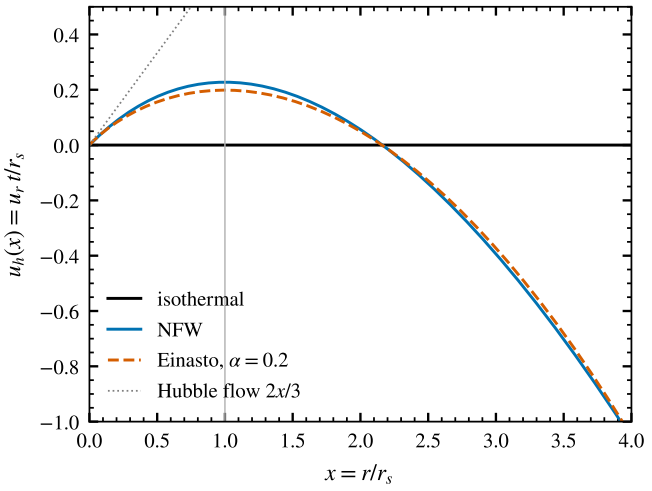


Figure 2. Mean radial velocity inside a self-similarly growing halo, $u_h = u_r\, t/r_s$ (equation 11), for the isothermal, NFW ($c = 4$) and Einasto ($\alpha = 0.2$) profiles. Haloes that deform flow outward in the core and inward beyond x_0; the flow is fastest at the scale radius $x = 1$. The dotted line is the Hubble flow, $2x/3$ in these units.

Fig. 2 shows $u_h(x)$ for the three profiles: outflow in the core, inflow outside x_0. The scale radius thus appears as the radius of fastest flow, a length that a scale-free isothermal profile does not have. In terms of the flow, the logarithmic slope of the density (equation 8) is

$$\frac{\partial \ln \rho_h}{\partial \ln x} = \frac{u'_h}{1 - u_h/x} - 2, \tag{12}$$

so a profile can be built from a model of the flow, and an inner flow $u_h = \gamma_h x$ gives an inner power law $\rho_h \propto r^{(3\gamma_h-2)/(1-\gamma_h)}$: r^{-2} for $\gamma_h = 0$, r^{-1} for $\gamma_h = 1/2$ and a core for $\gamma_h = 2/3$. The slower the deformation at the centre, the steeper the cusp.

Peculiar flow and stable clustering. The Hubble flow is $Hr = (2/3)x r_s/t$, so the peculiar radial velocity is $u_h - 2x/3 = x/3 - F/F'$. For the isothermal sphere it is $-Hr$: the halo neither expands nor contracts in physical coordinates, as in the stable clustering hypothesis [50]. A growing NFW or Einasto halo instead expands near its centre, with a total radial velocity $\gamma_h r/t$ there.

Angle of incidence. If the tangential speed of a particle equals the local circular velocity v_c, the angle between its peculiar velocity

and its position vector follows from $\cot\theta = u_p/(2\pi c\, v_c/v_{\rm cir})$. At the edge $\cot\theta = (1/3 - 1/\alpha_h)/(2\pi)$: $\theta = 96.1^\circ$ for $\alpha_h = 1$ and 98.4° for $\alpha_h = 0.79$ (NFW, $c = 4$). Matter falls in almost tangentially, slightly inward of a circle; $\theta \to 90^\circ$ at the centre.

A polynomial flow. The flow can be modelled directly. A cubic $u_h = \gamma_h x + a_2 x^2 + a_3 x^3$ with maximum at $x = 1$ and the edge value $c(1 - 1/\alpha_h)$ has

$$a_2 = -\frac{(c^2-3)\gamma_h + 3 - 3/\alpha_h}{2c^2 - 3c}, \qquad a_3 = \frac{(c-2)\gamma_h + 2 - 2/\alpha_h}{2c^2 - 3c}, \tag{13}$$

and equation (11) then gives F in closed form. The quadratic case ($a_3 = 0$) requires $\gamma_h = (1 - 1/\alpha_h)/(1 - c/2)$ and has $x_0 = 2$. Three numbers, the central deformation rate γ_h, the edge deformation α_h and the concentration c, thus describe a profile.

4 MOMENTUM, DISPERSION AND ENERGY

4.1 Radial momentum and a preferred concentration

The radial momentum and kinetic energy of the flow, in units of $m_h v_{\rm cir}$ and $m_h v_{\rm cir}^2$, are

$$\lambda_{Lr} = \frac{1}{m_h v_{\rm cir}} \int_0^{r_h} u_r \,{\rm d}m_r = \frac{cF(c) - 2\int_0^c F\,{\rm d}x}{2\pi c F(c)}, \tag{14}$$

$$\lambda_{Kr} = \frac{1}{2m_h v_{\rm cir}^2} \int_0^{r_h} u_r^2 \,{\rm d}m_r = \frac{c^2 F(c) - 4\int_0^c xF\,{\rm d}x + \int_0^c F^2/F'\,{\rm d}x}{8\pi^2 c^2 F(c)}. \tag{15}$$

Outflow in the core and inflow outside cancel at one concentration. For NFW, $\lambda_{Lr} = 0$ at

$$\frac{c(4+3c)}{1+c} = (4+c)\ln(1+c) \quad \Rightarrow \quad c_0 = 3.48 \tag{16}$$

(Fig. 3). A halo with $c = c_0$ grows with no net radial momentum; at larger c the inflow wins. The radial kinetic energy is tiny, $\lambda_{Kr}(c_0) = 6.8\times10^{-5}$, close to its minimum 4.7×10^{-5} at $c = 2.85$. Values of $c \approx 4$ are found for fast-accreting massive haloes in simulations [20, 22]; equation (16) offers a kinematic reason for a limiting concentration of this order.

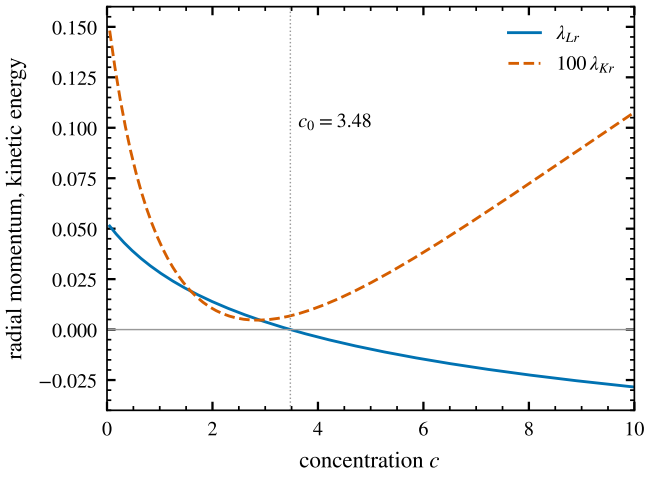


Figure 3. Radial momentum λ_{Lr} and radial kinetic energy λ_{Kr} (times 100) of a growing NFW halo against concentration (equations 14, 15). The momentum vanishes at $c_0 = 3.48$, near the minimum of the kinetic energy.

4.2 Velocity dispersion with radial flow

The radial Jeans equation of a spherical, non-rotating halo with isotropic dispersion σ_r^2 is

$$\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{1}{\rho_h}\frac{\partial(\rho_h\sigma_r^2)}{\partial r} = -\frac{Gm_r}{r^2}. \tag{17}$$

Usually the left-hand flow terms are dropped. Here they are known: with $u_r = (r_s/t)u_h(x)$ and self-similar growth they add F^2F''/F'^3 to the equation. Integrating inward from $\rho_h\sigma_r^2 = 0$ at infinity gives the dispersion in units of $v_{\rm cir}^2$ as two parts,

$$\frac{\sigma_r^2}{v_{\rm cir}^2} = \sigma_{nr1}^2 + \sigma_{nr2}^2, \tag{18}$$

with the radial-flow part and the gravitational part

$$\sigma_{nr1}^2 = \frac{x^2}{4\pi^2c^2F'(x)}\int_x^\infty \frac{F^2}{y^2}\left(\frac{1}{F'}\right)'{\rm d}y,$$

$$\sigma_{nr2}^2 = \frac{cx^2}{F(c)F'(x)}\int_x^\infty \frac{FF'}{y^4}{\rm d}y.$$

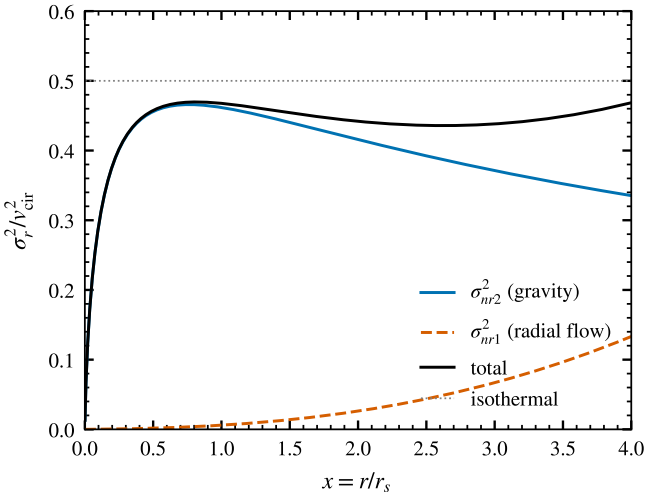


Figure 4. Radial velocity dispersion of a growing NFW halo ($c = 4$, equation 18): gravitational part, radial-flow part and total. The flow adds radial dispersion in the outer halo only.

The second part is the usual hydrostatic dispersion; for the isothermal sphere it is $1/2$. The first part comes from the flow. For NFW (Fig. 4; closed forms in Appendix B) it is negligible in the core and grows outward, where it is more than a quarter of the total at $x = 4$: the flow heats the outer halo radially. Near the centre the gravitational part behaves as $-(c/2F(c))\,x\ln x$, and the general power laws for an inner flow $\gamma_h x$ are $F \propto x^{1/(1-\gamma_h)}$, $\rho_h \propto x^{(3\gamma_h-2)/(1-\gamma_h)}$, $v_c^2 \propto x^{\gamma_h/(1-\gamma_h)}$, and $\sigma_r^2 \propto x^{\gamma_h/(1-\gamma_h)}$ for $\gamma_h < 1/2$ or $x^{(2-3\gamma_h)/(1-\gamma_h)}$ for $1/2 < \gamma_h < 2/3$.

4.3 Virial balance of a growing halo

Multiplying the continuity equation by u_r, adding ρ_h times equation (17), multiplying by $4\pi r^3$ and integrating from 0 to $r_h(t)$ gives an exact virial theorem for a growing, non-rotating, isotropic halo.

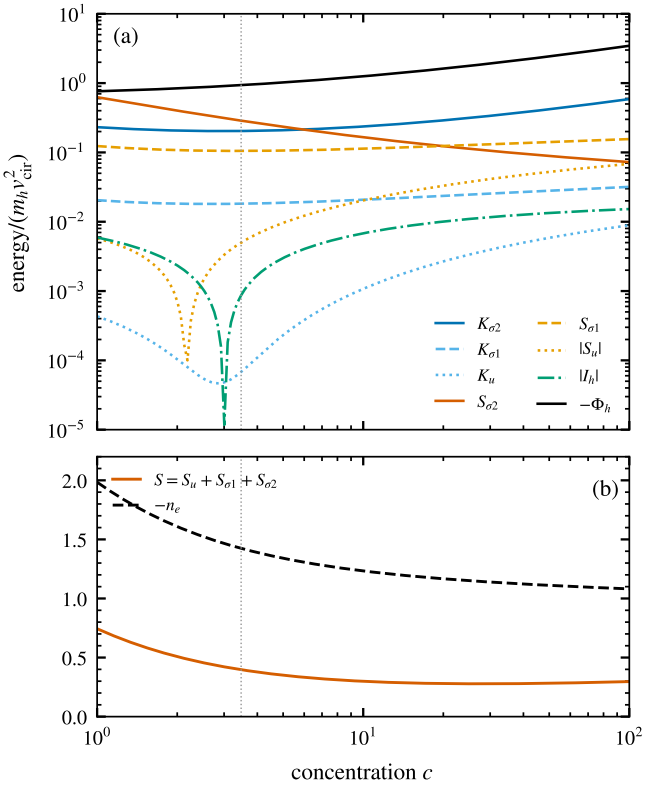


Figure 5. Virial balance of a growing NFW halo (equation 19) against concentration. (a) Kinetic (K), surface (S), inertial (I_h) and potential (Φ_h) terms in units of $m_h v_{\rm cir}^2$; $|S_u|$ and $|I_h|$ change sign at $c = 2.16$ and $c = 3$. (b) Total surface term and effective virial exponent; the dotted line marks c_0.

In units of $m_h v_{\rm cir}^2$,

$$I_h + S_u + S_\sigma - 2K_u - 6K_\sigma - \Phi_h = 0. \tag{19}$$

Here $\Phi_h = -(c/F(c)^2)\int_0^c FF'/x\,{\rm d}x$ is the potential energy, $K_u = \lambda_{Kr}$ the kinetic energy of the mean flow, $K_\sigma = K_{\sigma 1} + K_{\sigma 2}$ the kinetic energy of the radial dispersion (so $6K_\sigma$ is twice the total random kinetic energy for isotropy), I_h the rate of change of the first radial moment $G_h = \int 4\pi r^3 \rho_h u_r\,{\rm d}r$, and $S_u = (1/\alpha_h - 1)/(4\pi^2)$ and $S_\sigma = S_{\sigma 1} + S_{\sigma 2}$ the surface terms from the flow and from the pressure $\rho_h \sigma_r^2$ at the moving edge. The static virial theorem, $2K = -\Phi$, keeps only $K_{\sigma 2}$ and Φ_h.

Fig. 5 shows all terms for NFW. At c_0 the surface terms are $S_u = 0.005$, $S_{\sigma 1} = 0.105$ and $S_{\sigma 2} = 0.287$; the kinetic terms are $K_u = 6.8\times10^{-5}$, $K_{\sigma 1} = 0.018$ and $K_{\sigma 2} = 0.204$; $\Phi_h = -0.935$ and $I_h = -8.7\times10^{-4}$ (zero at $c = 3$). The flow carries little kinetic energy, but its surface term $S_u + S_{\sigma 1}$ is about 40 per cent of $S_{\sigma 2}$: it acts mainly at the edge, where the flow is fastest relative to the density. The total surface energy,

$$S \equiv S_u + S_{\sigma 1} + S_{\sigma 2} = 2(K_u + 3K_{\sigma 1} + 3K_{\sigma 2}) + \Phi_h - I_h, \tag{20}$$

is the energy needed to keep an expanding edge; $S = 0.40$ at c_0 (0.38 at $c = 4$; 1/2 for the isothermal sphere). The pressure jump across the edge is $\Delta P_h = S\bar\rho_h v_{\rm cir}^2/3 = 0.13\,\bar\rho_h v_{\rm cir}^2$.

Effective exponent of gravity. For a potential $V \propto r^n$ the static virial theorem reads $2K = n\Phi$. A growing halo satisfies the same form with an effective exponent

$$n_e = \frac{2(K_u + 3K_{\sigma 1} + 3K_{\sigma 2})}{\Phi_h} = \frac{S + I_h}{\Phi_h} - 1, \tag{21}$$

which is $n_e = -1.42$ at c_0 (-1.39 at $c = 4$; -1.5 for the isothermal sphere) instead of -1. The surface energy of growth makes a halo look as if its gravity were steeper than $1/r$. For comparison, the whole simulated system, all particles rather than one halo, has $n_e \approx -1.37$ at $z = 0$, which follows from the cosmic energy equation and the measured growth of the kinetic energy [51]; the two numbers describe different objects.

Surface tension. Defining a surface tension $S_{th} = Sm_h v_{\rm cir}^2/(2A_h)$ with $A_h = 4\pi r_h^2$ (the factor 1/2 is a convention that gives the Young–Laplace form $\Delta P_h = 2S_{th}/r_h$), dimensional analysis with G, the surface density $\rho_{\rm sur}$ and r_h gives $S_{th} = \alpha_{st} G\rho_{\rm sur}^2 r_h$ and $\rho_{\rm sur} = \sqrt{S/(8\pi\alpha_{st})}\,m_h/r_h^2$. For self-similar growth ($m_h \propto r_h$), $S_{th} \propto r_h^{-1}$, $\rho_{\rm sur} \propto r_h^{-1}$, and the thickness of the surface layer, $r_p = \rho_{\rm sur}/\rho_h(r_h)$, is proportional to r_h for any λ and τ_0. The halo radius therefore grows in steps proportional to itself, which is the starting point of the next section.

5 STOCHASTIC GROWTH AND THE HALO PROFILE

5.1 Halo size as geometric Brownian motion

Mergers arrive at random times. If the merger count in a time ${\rm d}t$ is Poisson with mean ${\rm d}t/\tau_g$, the mass grows on average at m_p/τ_g with fluctuations of relative size $(\tau_g/{\rm d}t)^{1/2}$; in the diffusion limit this is a Gaussian noise added to equation (4). Because the surface step r_p is proportional to r_h, the noise of the radius is multiplicative. In terms of $\ln a$,

$$\frac{{\rm d}r_h}{{\rm d}\ln a} = \frac{3}{2} b_{rh} r_h + r_h \xi_{rh}, \tag{22}$$

with $\langle \xi_{rh}(\ln a)\,\xi_{rh}(\ln a')\rangle = 2D_{rh}\,\delta(\ln a - \ln a')$, with $b_{rh} = (9 - 2\tau_0 - 6\lambda)/(9(1-\lambda)) = 1$ for self-similar growth and a dimensionless diffusion coefficient D_{rh}. If the noise comes from fluctuations of the edge at the radial dispersion, $\sigma_r(r_h) = \alpha_{rh} v_{\rm cir} = 3\pi\alpha_{rh} H r_h$, then $D_{rh} = (3\pi\alpha_{rh}\beta_{rh})^2/2$, where β_{rh} is the fraction of that motion that changes the radius. Equation (22) is geometric Brownian motion. Read in the Itô sense, its solution starting from r_{h0} at t_i is lognormal, with

$$\langle \ln(r_h/r_{h0})\rangle = \left(b_{rh} - \tfrac{2}{3}D_{rh}\right)\ln\frac{t}{t_i}, \qquad {\rm Var}\ln r_h = \tfrac{4}{3}D_{rh}\ln\frac{t}{t_i}, \tag{23}$$

and moments $\langle r_h^m\rangle = r_{h0}^m (t/t_i)^{mb_{rh} + 2m(m-1)D_{rh}/3}$. The mean radius grows as $t^{b_{rh}}$, the median as $t^{b_{rh} - 2D_{rh}/3}$ and the mode as $t^{b_{rh} - 2D_{rh}}$. Halo mass, proportional to r_h, is lognormal too.

5.2 Particles in a randomly growing halo

A particle at $r_t = x_t r_s(t)$ moves because the halo grows (r_s changes randomly) and because it moves relative to the halo (x_t changes). Following Nelson's stochastic mechanics [52], a diffusion has a forward and a backward drift; their mean is the current velocity, here the mean flow u_h, and half their difference is the osmotic velocity u_h^*, which in Brownian motion balances the diffusion flux [53]. With equation (22) the position obeys

$$\frac{{\rm d}r_t}{{\rm d}t} = \frac{r_s}{t}\left[u_h(x_t) \pm u_h^*(x_t)\right] + \sigma(x_t)\, r_s H \xi_{rh}(t), \tag{24}$$

with $+$ forward and $-$ backward in time and multiplicative noise $\sigma(x) = x$ inherited from the halo size. Adding and subtracting the two Fokker–Planck equations of the mass distribution per unit x,

$P_r(x)$ (independent of time for self-similar growth because $m_h \propto r_s$), gives

$$x\frac{\partial P_r}{\partial x} = \frac{\partial(u_h P_r)}{\partial x}, \qquad u_h^* = d_r\,\sigma^2\frac{\partial \ln(\sigma^2 P_r)}{\partial x}, \qquad d_r = \frac{2}{3}D_{rh}. \quad (25)$$

The first equation integrates back to equation (11) with $F = \int_0^x P_r$; the second ties the osmotic velocity to the profile. Eliminating P_r gives, for $\sigma = x$,

$$u_h^* = d_r\left[\frac{x^2 u_h'}{x-u_h} + 2x\right], \quad (26)$$

and near the centre $u_h^* \simeq \gamma_r x$ with $\gamma_r = d_r(2-\gamma_h)/(1-\gamma_h)$. One more relation between u_h and u_h^* closes the problem. We do not have it from first principles (Appendix C discusses a non-local candidate); instead we model the osmotic flow. It rises as $\gamma_r x$, takes the value $2d_r$ at $x = 1$ (where $\partial \ln P_r/\partial \ln x = 0$) and vanishes where the density slope is -4. The simplest form with these properties is $u_h^* = \gamma_r x - \beta_r x^{1+\alpha_r}$ with $\beta_r = \gamma_r - 2d_r$. Equation (25) then gives a normalized generalized gamma distribution,

$$P_r(x) = \frac{b_r^{a_r}}{\Gamma(a_r)(a_r-b_r)}\,x^{\frac{b_r}{a_r-b_r}}\exp\left(-b_r x^{\frac{1}{a_r-b_r}}\right),$$
$$F_r(x) = \frac{\gamma\left(a_r, b_r x^{1/(a_r-b_r)}\right)}{\Gamma(a_r)}, \quad (27)$$

with $a_r = (\gamma_r - d_r)/(\alpha_r d_r)$, $b_r = \beta_r/(\alpha_r d_r)$, $\alpha_r = 1/(a_r - b_r)$ and γ the lower incomplete gamma function. We call it the cascade profile. Its density, $\rho_h = m_h P_r/(4\pi r_s^3 x^2)$, has the inner slope $(3b_r - 2a_r)/(a_r - b_r)$; the central deformation rate is $\gamma_h = b_r/a_r$. For $a_r/b_r = 3/2$ ($\gamma_h = 2/3$) the density is exactly the Einasto profile with $\alpha = 2/b_r$. The moments are $\langle x^k\rangle = \Gamma(a_r + k/\alpha_r)/[b_r^{k/\alpha_r}\Gamma(a_r)]$, so the rms radius is

$$r_g = r_s\left[\frac{\Gamma(3a_r - 2b_r)}{b_r^{2(a_r-b_r)}\Gamma(a_r)}\right]^{1/2}. \quad (28)$$

Equation (27) describes the distribution of particles over haloes of all sizes, extending to infinity; it is meant for an ensemble of haloes, such as a composite halo, rather than for one truncated halo.

6 COMPARISON WITH THE SIMULATION

6.1 Cumulative profiles

Fig. 6 shows $F_r(r)$ of five composite haloes at $z = 0$ with the cascade profile (27) fitted by least squares in $\log_{10} F_r$ over $0.01 \leqslant r \leqslant 2\,\mathrm{Mpc}/h$ (free parameters a_r, b_r, r_s). The model follows the data over four decades of F_r: for the 49 groups with $4 \leqslant n_p \leqslant 100$ the rms residual is 0.006 dex (median) and at most 0.013 dex. At small radius $F_r \propto r^3$, a constant density, but the local logarithmic slope of the density is -0.13 between 0.01 and 0.02 Mpc/h, -0.16 between 0.02 and 0.04, -0.52 between 0.04 and 0.08 and -1.27 between 0.08 and 0.16 Mpc/h (groups $20 \leqslant n_p \leqslant 100$). The flat part lies inside about one softening length, so these data cannot decide whether the haloes have physical cores.

6.2 Parameters and length scales

Fig. 7(a) shows the fitted parameters against n_p. Both a_r and b_r increase with halo mass (b_r from 1.70 at $n_p = 4$ to 3.80 at $n_p = 100$, with scatter up to 7.2 for single groups, because a_r and b_r are strongly correlated), while their ratio stays almost constant, $a_r/b_r = 1.41 \pm 0.05$ (range 1.29–1.52). The exponent $\alpha_r = 1/(a_r - b_r)$ falls from 1.25 to 0.64, far above the $\alpha \approx 0.2$ of Einasto fits to single haloes [5]; composite haloes mix sizes and are more extended. The ratio 3/2 that the model needs for a finite central density, and that the central Hubble flow of Section 6.3 would imply, is not reached at $z = 0$: $a_r/b_r = 1.41$ corresponds to $\gamma_h \approx 0.71$ and a formally rising inner density, $\rho_h \propto r^{+0.49}$, inside the softening length. At higher redshift the ratio is closer to 3/2 (Table 2).

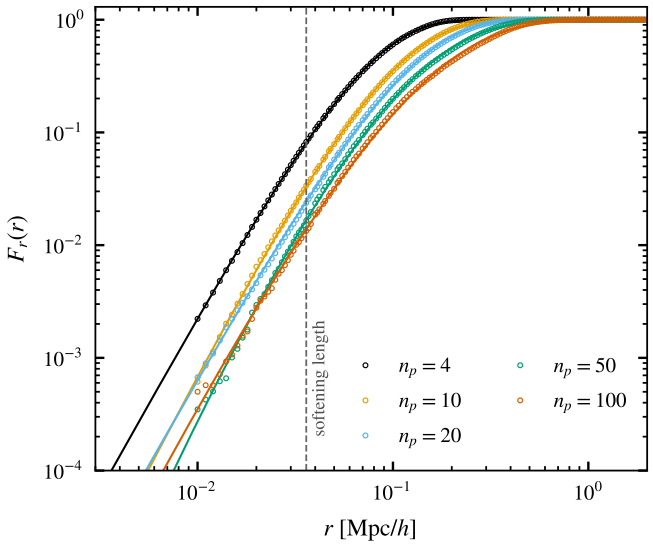


Figure 6. Cumulative mass fraction $F_r(r)$ of composite haloes with n_p particles at $z = 0$ (symbols; equation 1) and fits of the cascade profile (lines; equation 27). The profile fits all groups; the r^3 part at small radius lies inside the softening length (dashed).

Fig. 7(b) compares three radii: the fitted scale radius r_s, the rms radius r_g (computed directly from the particles) and the radius r_{h178} inside which the mean density of the composite halo is $18\pi^2\bar\rho$. They grow as $n_p^0.18$, $n_p^0.31$ and $n_p^0.34$; $r_{h178} \propto n_p^{1/3}$ as expected for a fixed mean density, while r_s grows more slowly, so the composite concentration r_{h178}/r_s rises from 2.08 to 3.28. The scale radius is only 2.0–4.2 softening lengths, so the inner fit is close to the resolution limit.

Table 2 extends the fits to larger haloes and higher redshift. The ratio a_r/b_r lies between 1.35 and 1.54 in all bins. At fixed n_p the comoving radii shrink slowly with time, so haloes of a given mass become more compact.

6.3 Mean radial flow and velocity dispersion

The kinematic result (11) predicts the flow from the profile alone: inserting the fitted F_r and $t = 2/(3H_0)$, a self-similarly growing composite halo should flow outward inside $x_0 \approx 1.5$–2 and inward beyond, with a total radial velocity (peculiar plus Hubble) of $+0.53$ to $+0.62\,Hr_s$ at the scale radius. Fig. 8(a) compares this with the measured mean total radial velocity of composite haloes, in units of the local Hubble velocity Hr, with errors from a jackknife over the n_p groups. The measured flow is inward at every resolved radius and roughly proportional to r: averaged over $2\,l_{\rm soft} < r < r_g$ it is $\langle u_r\rangle = -(0.39 \pm 0.02)Hr$ for $20 \leqslant n_p < 50$, $-(0.56 \pm 0.02)Hr$ for $50 \leqslant n_p < 100$, $-(0.73 \pm 0.02)Hr$ for $100 \leqslant n_p < 300$, $-(1.24 \pm 0.05)Hr$ for $300 \leqslant n_p < 1000$ and $-(1.10 \pm 0.13)Hr$ above. There

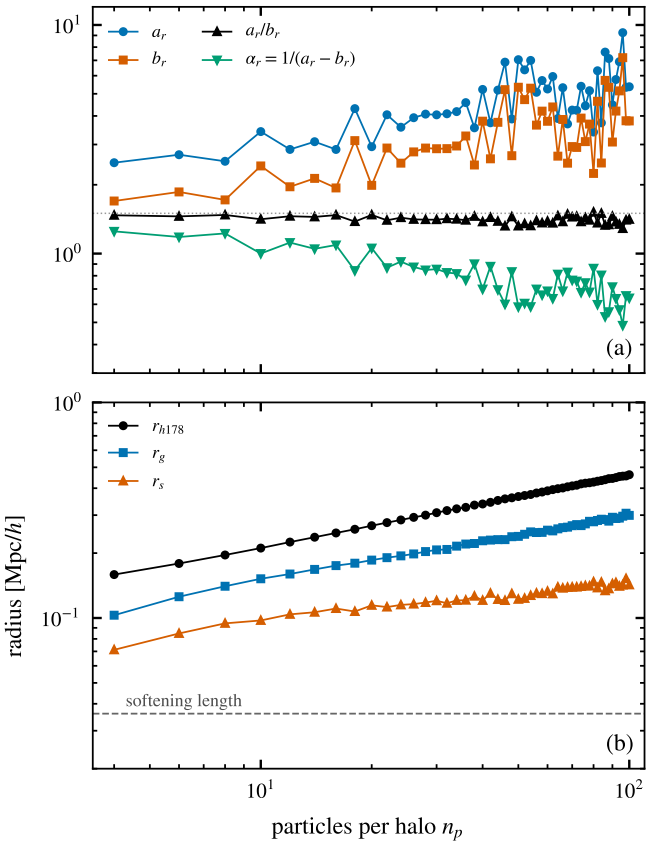


Figure 7. Fitted parameters of composite haloes at $z = 0$ against particle number. (a) a_r, b_r, a_r/b_r and $\alpha_r = 1/(a_r - b_r)$; the ratio stays near 1.4, below the 3/2 (dotted) of a finite central density. (b) Scale radius r_s, rms radius r_g and radius r_{h178} of mean density $178\pi^2\bar{\rho}$, with the softening length.

is no outflowing core. Near the centre the total flow is consistent with zero, as in stable clustering, not with the Hubble flow $+Hr$ that a central core would require. Outside the halo the measured inflow is much weaker than the steep inflow that equation (11) gives for an exponentially falling profile.

Fig. 8(b) tests the dispersion. The isotropic hydrostatic Jeans equation (the gravitational part of equation 18, with the fitted F_r and $m_h = n_p m_p$) underestimates the measured radial dispersion: σ_r^2 exceeds it by 1.17–1.55 times at the scale radius and by up to 1.90 times at r_g, the excess growing outward and with halo mass. The tangential dispersion exceeds it by only 1.02–1.52 times, so the haloes are radially anisotropic, $\beta_h = 1 - \sigma_t^2/\sigma_r^2 = 0.12$–$0.21$. An outward rise of radial dispersion is what the flow term of equation (18) produces (Fig. 4), but the measured flow has the opposite sign in the core; the extra dispersion more likely reflects ongoing radial infall and the anisotropy that the isotropic equation omits.

6.4 The core in hydrostatic equilibrium

Although the measured cores are not resolved, the cascade profile has a simple core when $a_r = 3b_r/2$: $\rho_h \simeq \rho_h(0)(1 - b_r x^{2/b_r})$ with $\rho_h(0) = 2b_r^{3b_r/2-1}/\Gamma(3b_r/2)$ in units of $m_h/(4\pi r_s^3)$. In isotropic hydrostatic equilibrium the pressure $p_h = \rho_h\sigma_r^2$ near the centre falls as $\Delta p_h = p_h(0) - p_h = \rho_h(0)^2 x^2/6$ (units $Gm_h/r_s \times m_h/4\pi r_s^3$), so

Table 2. Cascade-profile fits of composite haloes binned in n_p at four redshifts (fit on logarithmic bins, $0.01 \leqslant r \leqslant 2$ comoving Mpc/h; bins with at least 20,000 particles). Radii in comoving Mpc/h.

z	n_p	particles	a_r	b_r	a_r/b_r	r_s	r_g
0	4–9	1 647 414	2.78	1.91	1.46	0.080	0.122
0	10–19	868 980	3.26	2.28	1.43	0.105	0.167
0	20–49	1 161 156	3.85	2.70	1.43	0.120	0.211
0	50–99	911 025	4.80	3.41	1.41	0.137	0.269
0	100–299	1 420 010	7.04	5.18	1.36	0.163	0.363
0	300–999	1 154 195	8.48	6.29	1.35	0.225	0.548
0	$\geqslant 1000$	371 757	4.95	3.36	1.47	0.358	0.883
0.5	4–9	1 609 250	2.22	1.51	1.48	0.091	0.127
0.5	10–19	844 729	2.26	1.53	1.48	0.123	0.177
0.5	20–49	1 028 976	2.74	1.86	1.48	0.144	0.228
0.5	50–99	686 415	3.68	2.55	1.45	0.165	0.293
0.5	100–299	771 398	4.49	3.13	1.43	0.200	0.396
0.5	300–999	260 076	4.63	3.18	1.46	0.273	0.580
1	4–9	1 346 157	1.89	1.26	1.49	0.097	0.129
1	10–19	661 055	1.95	1.30	1.50	0.131	0.181
1	20–49	693 768	2.69	1.83	1.47	0.152	0.236
1	50–99	360 821	3.34	2.29	1.46	0.177	0.305
1	100–299	238 183	4.21	2.93	1.44	0.213	0.405
1	300–999	33 390	4.67	3.28	1.42	0.277	0.593
2	4–9	588 076	1.51	1.00	1.51	0.102	0.130
2	10–19	213 637	1.99	1.35	1.48	0.135	0.184
2	20–49	142 387	2.25	1.51	1.49	0.162	0.240
2	50–99	31 879	2.07	1.35	1.54	0.200	0.314

pressure and density differences obey a power law,

$$\Delta p_h = \frac{\rho_h(0)^{2-b_r}}{6\,b_r^{b_r}}\,(\Delta\rho_h)^{b_r}, \tag{29}$$

an equation of state for differences measured from the centre, where both pressure gradient and gravity vanish. The exponent of the core is b_r. Fig. 9(a) shows that equation (29) holds within 10 per cent up to $x = 0.32$ for $b_r = 5/3$. Integrating the hydrostatic equation to infinity gives the central values in Fig. 9(b): for $b_r = 5/3$, $\rho_h(0) = 3.24$, $\sigma_{r0}^2 = 0.147\,Gm_h/r_s$ and a core radius (where the parabolic pressure would vanish) $x_c = 0.52$; $x_c = 0.80$ and 0.20 for $b_r = 1$ and 3. These values follow from the profile alone; the radial flow, small compared with the dispersion in the data, is neglected.

7 DISCUSSION

What holds. Equations (10)–(12) are exact consequences of mass conservation for any halo that keeps the form (7); they turn a growth history into a velocity field and back. The vanishing radial momentum at $c_0 = 3.48$, the generalized virial balance (19) with its surface terms, and the geometric Brownian growth of the radius are exact within their assumptions and were checked numerically (Appendix B). The cascade profile, which contains the Einasto profile, describes the stacked mass profiles of the simulation to better than 0.013 dex over four decades of F_r, with a nearly constant ratio a_r/b_r.

What fails. Self-similar growth at fixed concentration does not describe the instantaneous flow in the simulated haloes. Their mean radial velocity is inward everywhere, $\langle u_r\rangle \approx -(0.4$–$1.2)Hr$, where the model predicts an outflowing core. The central Hubble flow that would make $a_r/b_r = 3/2$ and a core is not seen, and the cores in the stacked profiles are at the resolution limit. Three effects may contribute: composite haloes at fixed n_p are not the same haloes followed in time, so their flow mixes haloes that grow, merge and are

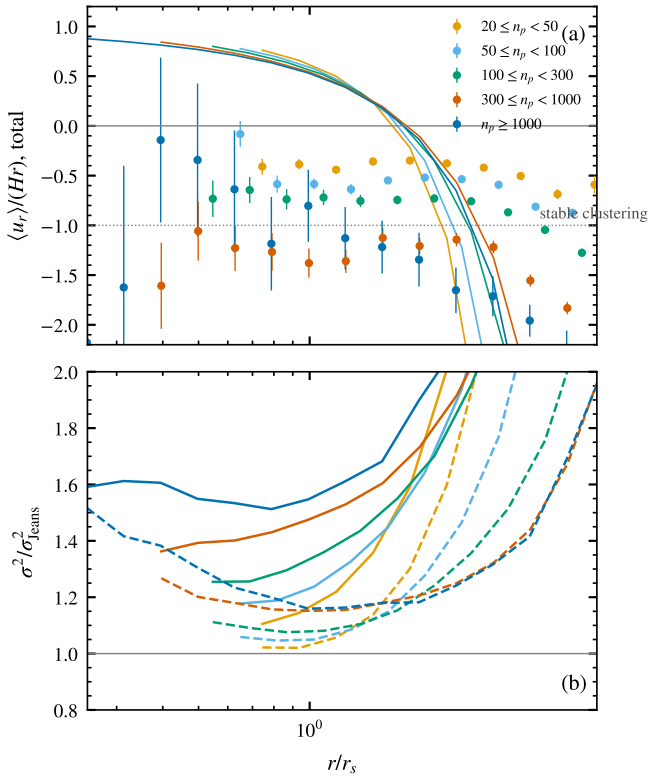


Figure 8. Kinematics of composite haloes at $z = 0$. (a) Mean total radial velocity (peculiar plus Hubble flow) in units of Hr against r/r_s (symbols, jackknife errors) and the prediction of self-similar growth from the fitted profile of each bin (lines). The measured flow is inward everywhere; the prediction is outward inside x_0. (b) Measured σ_r^2 (solid) and σ_t^2 (dashed) divided by the isotropic hydrostatic Jeans dispersion of the fitted profile. Radii below $2\,l_{\rm soft}$ are omitted.

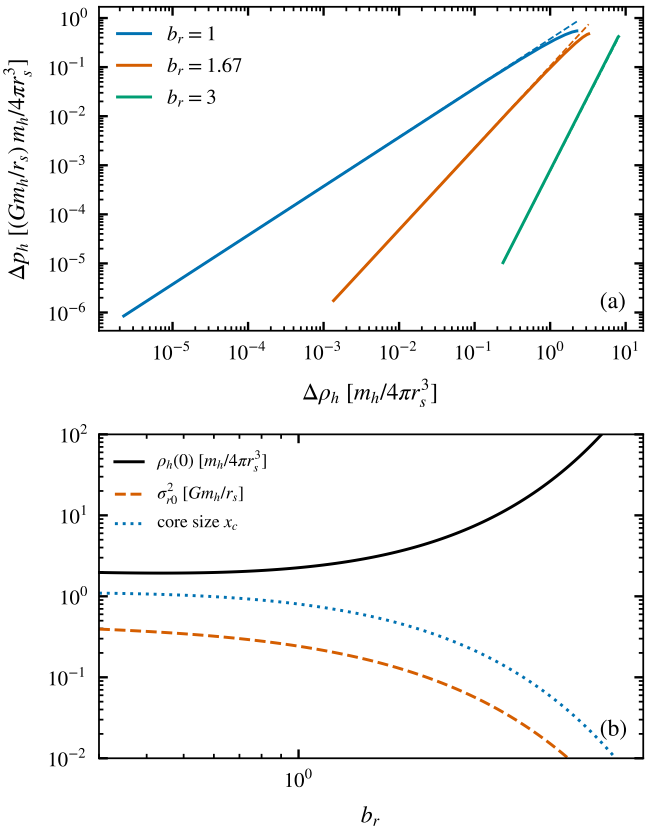


Figure 9. Core of the cascade profile with $a_r = 3b_r/2$ in isotropic hydrostatic equilibrium. (a) Pressure difference against density difference from the centre for three values of b_r (solid) and the core law (29) (dashed). (b) Central density, central radial dispersion and core radius against b_r.

stripped; friends-of-friends haloes include infalling matter near their edge; and at $z = 0$ the SCDM haloes grow more slowly than t (the total halo mass grows as about $a^{1/2}$ there [39]). Equation (10) also allows $\partial \ln m_h/\partial \ln t \neq \partial \ln r_s/\partial \ln t$. Fitting its two growth rates to the measured flow at constant c gives $\partial \ln r_s/\partial \ln t$ between -0.95 and -0.20 and $\partial \ln m_h/\partial \ln t$ between -0.15 and 0.05 for $n_p < 1000$ (0.66 above): the inward flow, proportional to r, is what a profile that contracts in physical size at nearly fixed mass would have.

Tests. (i) Follow individual haloes through snapshots and measure $\partial \ln m_h/\partial \ln t$, $\partial \ln r_s/\partial \ln t$ and $u_r(r)$ for the same objects; equation (10) must then hold exactly. (ii) Repeat the profile fits in a higher-resolution simulation: a physical core requires $a_r/b_r \to 3/2$ at radii well above the softening length. (iii) Compare the concentration of fast-growing haloes with $c_0 = 3.48$ in ΛCDM, where the growth rates differ. (iv) Measure the surface terms of equation (19) for single haloes; they predict $n_e \approx -1.42$ for $c \approx c_0$.

Open problems. The closure between mean and osmotic flow (Appendix C) is still a model, and the anisotropy of the dispersion, $\beta_h \approx 0.1$–0.2, should be added to equations (17) and (19).

8 CONCLUSIONS

(i) A halo that grows self-similarly (mass and radius proportional to time, fixed concentration) must contain a mean radial flow $u_r = (r_s/t)[x - F(x)/F'(x)]$, outward in the core and inward outside, fastest at the scale radius; the isothermal sphere is the only profile without flow. The deformation of the halo by accreted mass, $\alpha_h = cF'(c)/F(c)$, links the concentration to the growth.

(ii) For NFW the radial momentum of this flow vanishes at $c_0 = 3.48$, close to the minimum of its kinetic energy.

(iii) Growth adds surface and kinetic terms to the virial theorem; at c_0 the surface energy is $S = 0.40\, m_h v_{\rm cir}^2$ and the effective virial exponent is $n_e = -1.42$.

(iv) A halo radius that grows by random, size-proportional steps follows geometric Brownian motion and a lognormal distribution. Particles in such a halo obey forward and backward Fokker–Planck equations; a simple osmotic flow gives the cascade profile $F_r = \gamma(a_r, b_r x^{1/(a_r - b_r)})/\Gamma(a_r)$, which contains the Einasto profile.

(v) In the SCDM simulation the cascade profile fits composite haloes of 4–6326 particles with a median rms residual of 0.006 dex and $a_r/b_r = 1.41 \pm 0.05$ at $z = 0$; the apparent cores lie within one softening length.

(vi) The measured mean radial velocity of the composite haloes is inward at all resolved radii, $-(0.39$–$1.24)Hr$, and their radial dispersion exceeds the isotropic Jeans value. Self-similar growth therefore describes the profile shapes but not the present kinematics of these haloes.

DATA AVAILABILITY

The halo-based and correlation-based statistics of the simulation are available on Zenodo [46, 47], with presentation slides [54] and all data files [48]. The per-particle distances of the composite haloes, the binned velocity profiles, and the code that reproduces every figure and number of this paper are distributed with it.

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APPENDIX A: HALO DENSITY PROFILES

NFW [1]: $\rho_h = \rho_s/[x(1+x)^2]$, $F(x) = \ln(1+x) - x/(1+x)$, $\mathrm{d}\ln\rho_h/\mathrm{d}\ln r = -(1+3x)/(1+x)$; the mean density inside r is $\bar\rho_r = 3m_hF(x)/(4\pi r^3F(c))$, so $\rho_h = \bar\rho_r/[3F(x)(1+1/x)^2]$, and at the edge $\rho_h(r_h) = \bar\rho_h/[3F(c)(1+1/c)^2]$.

Power law: $\rho_h = (3-m)m_h(r/r_h)^{-m}/(4\pi r_h^3) = (3-m)\bar\rho_r/3$; at the edge $(3-m)m_h/(4\pi r_h^3)$. The local density is fixed by the mean density inside r, unlike NFW and Einasto.

Einasto [55]: $\rho_h = \rho_s e^{2/\alpha}\exp[-(2/\alpha)x^\alpha]$, $\mathrm{d}\ln\rho_h/\mathrm{d}\ln r = -2x^\alpha$, and

$$m_r = 4\pi\rho_s r_s^3\frac{e^{2/\alpha}}{\alpha}\left(\frac{\alpha}{2}\right)^{3/\alpha}\gamma\left(\frac{3}{\alpha},\frac{2x^\alpha}{\alpha}\right). \tag{A1}$$

Requiring an Einasto and an NFW halo of the same concentration to have the same density at r_h gives

$$cF(c)(1+c)^2 = \frac{\gamma(3/\alpha, 2c^\alpha/\alpha)}{\alpha(2/\alpha)^{3/\alpha}e^{-2c^\alpha/\alpha}}, \tag{A2}$$

whose solution (Fig. A1) decreases from $\alpha = 0.201$ at $c = 4$ to 0.171 at $c = 10$: more massive, less concentrated haloes have larger α, as in simulations [5, 56].

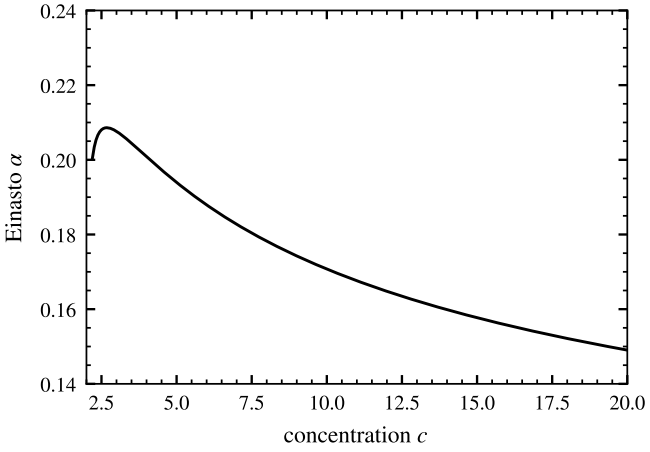


Figure A1. Shape parameter α of the Einasto profile that has the same concentration and edge density as an NFW profile (equation A2).

APPENDIX B: CLOSED FORMS FOR NFW

All closed forms below were checked against direct quadrature to relative accuracy 10^{-8} or better (code/checks in the accompanying package). With $L = \ln(1+c)$ and $F(c) = L - c/(1+c)$:

$$\lambda_{Kr} = \frac{1}{32\pi^2c^2F(c)}\Big[\frac{c(13c^2+7c-10)}{1+c} + 8\zeta(3) - 8\,\mathrm{Li}_3(1+c) + 2L\big(5-10c-5c^2+(3+4c+c^2+2\ln(-c))L\big) + 8L\,\mathrm{Li}_2(1+c)\Big], \tag{B1}$$

$$I_h = \frac{(c-3)\left[5c^2+6c-2(1+c)(c+3)L\right]}{8\pi^2c^2(1+c)F(c)}, \tag{B2}$$

$$S_u = \frac{(1+c)^2F(c)-c^2}{4\pi^2c^2}, \tag{B3}$$

$$\Phi_h = -\frac{c\left[c(2+c)-2(1+c)L\right]}{2F(c)^2(1+c)^2}, \tag{B4}$$

where the polylogarithms are taken on the branch $\mathrm{Im}\,\mathrm{Li}_2(1+c) = -\pi L$, $\mathrm{Im}\,\mathrm{Li}_3(1+c) = -\pi L^2/2$ with $\ln(-c) = \ln c + i\pi$, so that the imaginary parts cancel. The two parts of the dispersion (equation 18) are, with $\ell = \ln(1+x)$,

$$\sigma_{nr1}^2 = -\frac{(1+x)^2}{36\pi^2x^2c^2}\Big\{3x^2-4\pi^2x^3+12x^3\mathrm{Li}_2(1+x) - \left[3(1+x)^2(2x-1)\ell+6x-12x^2 - 12x^3\ln x - 12i\pi x^3\right]\ell\Big\}, \tag{B5}$$

$$\sigma_{nr2}^2 = \frac{c\,\ell}{2xF(c)} + \frac{c}{2F(c)}\Big\{-1-9x-7x^2+(-2-8x-4x^2+x^3)\ell + \left[\pi^2+6\mathrm{Li}_2(-x)-\ln x+3\ell^2\right]x(1+x)^2\Big\}, \tag{B6}$$

with $\sigma_{nr1}^2 \simeq (1/18 - 1/3\pi^2)x/c^2$ and $\sigma_{nr2}^2 \simeq -(c/2F(c))x\ln x$ as $x \to 0$.

APPENDIX C: A NON-LOCAL CLOSURE

In Brownian motion the current and osmotic velocities balance, $u_h = -u_h^*$, because the drag force is local; this gives the diffusion equation and the Einstein relation [53]. Gravity is long-ranged, so the osmotic velocity at radius r may depend on the flow inside r. A simple choice makes it proportional to the mean radial velocity of the matter inside

r,

$$u_h^* = \lambda_h \left[x - \frac{2}{F(x)} \int_0^x F(y)\,\mathrm{d}y \right]. \tag{C1}$$

Combined with equation (26) for $\sigma = x$ this gives a third-order equation for the profile,

$$F''' + \frac{F''F'}{F} + \frac{2F''}{x} - \frac{F'^2}{F'} + \left(2 + \frac{1}{d_h}\right)\frac{F'}{x^2} + \left(2 - \frac{1}{d_h}\right)\frac{F'^2}{xF} = 0, \tag{C2}$$

with $d_h = d_r/\lambda_h$, $F(0) = 0$, $F(\infty) = 1$ and $F''(1) = 0$. Power-law solutions $F \propto x^{n_1}$ at small x have $n_1 = \frac{1}{2}[(1/d_h - 2) \pm \sqrt{(1/d_h)(1/d_h - 8)}]$, so $d_h = 1/8$ gives $n_1 = 3$, a core; at large x, $F' \propto x^{-(2+1/d_h)}$, i.e. x^{-10} for $d_h = 1/8$. Whether this closure describes simulated haloes remains to be tested.